MAJAHA MOYO

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PROFESSIONAL PROFILE

A motivated and technically skilled **Computer Systems Engineering graduate** with a strong foundation in software development, systems integration, ICT infrastructure, networking, cloud technologies, and embedded systems. Experienced in developing and integrating software solutions using **Python, Flask, JavaScript, REST APIs, PostgreSQL, Docker, and database technologies**. Possesses practical knowledge of designing, implementing, and troubleshooting technology solutions across software and infrastructure environments.

Demonstrates strong analytical and problem-solving abilities, attention to detail, adaptability, and effective teamwork. Passionate about leveraging technology to develop **reliable, scalable, secure, and innovative solutions** that address real-world business and technical challenges.

Portfolio: https://majaha-portfolio.netlify.app/

TECHNICAL SKILLS

Software Development: Python, Flask, JavaScript, SQL, REST API Development, System Integration, Git & GitHub

Web & Database: Flask-SQLAlchemy, PostgreSQL, Database Design, API Integration, Backend Development

Infrastructure & Networking: TCP/IP, Routing & Switching, Linux Administration, Enterprise Networking, PABX & VoIP, Cloud Services (NaaS, IaaS, DaaS)

Security & ICT Systems: CCTV Systems, Video Management Systems, Access Control, Intrusion Detection, Data Protection

Engineering & Embedded: PLC Programming, Robotics & Automation (ROS), Digital Electronics, Arduino

Professional Skills: Problem Solving, Team Collaboration, Leadership, Agile Development, Project Documentation

WORK EXPERIENCE

Software Developer Intern · Firmlinx (Pty) Ltd

March 2026 – Present | Emalahleni, Mpumalanga, South Africa · On-site

- Responsible for system integration across internal and external platforms, connecting frontend and backend systems using Python (Flask) and JavaScript.
- Develop, maintain, and optimize APIs to ensure seamless data flow between services and third-party applications.
- Collaborate with cross-functional teams to design scalable architectures, troubleshoot integration issues, and enhance system performance.
- Ensure secure communication between systems while improving overall application reliability and user experience.
- Participate in backend development, database integration, testing, deployment, and system maintenance activities.
- Contribute to business applications involving ticket management, integrations, reporting, notifications, and workflow automation.

Technologies: *Python, Flask, JavaScript, SQL, PostgreSQL, REST APIs, Git, System Integration*

IT Technician Intern · Mctrends Systems

August 2025 –February 2026 | South Africa · On-site

- Implemented and maintained integrated security and ICT infrastructure solutions.
- Worked with Advanced CCTV Systems, Video Management Solutions, Perimeter Protection, Access Control Systems, and Intrusion Detection Systems.
- Supported Remote/Offsite Monitoring and Security Systems Integration projects.
- Assisted with Enterprise Networking, Server and Data Centre infrastructure, and Data Protection initiatives.
- Supported PABX & VoIP deployments, Cloud Services (NaaS, IaaS, DaaS), and Data Engineering projects.
- Contributed to infrastructure troubleshooting, maintenance, and technical support activities.

Technologies: *Networking, CCTV Systems, Access Control, VoIP, Cloud Services*

Auto Electrical Assistant · Wellzz Auto

November 2019 – June 2022 | Randburg, South Africa

- Diagnosed and repaired electrical issues in motor vehicles.
- Performed battery testing, alternator and starter motor repairs, and wiring installations.
- Used diagnostic tools and software for vehicle fault-finding.
- Maintained tool inventory and ensured workshop safety compliance.
- Gained practical insight into vehicle electrical systems, teamwork, and workshop operations.

PROJECTS

Smart Irrigation and Water monitoring system · Final Year Project

2024

- Designed and developed an IoT-based smart irrigation system to automate irrigation and improve water management through real-time environmental monitoring.
- Integrated soil moisture, temperature, and humidity sensors with a microcontroller to collect and process environmental data.
- Implemented automated irrigation control that activates and deactivates a water pump based on predefined soil moisture thresholds.
- Developed remote monitoring and control functionality, enabling users to monitor system conditions and manage irrigation remotely.
- Implemented a web-based monitoring interface for displaying sensor readings and system status in real time.
- Applied embedded systems, IoT, sensor integration, automation, and software-hardware communication principles to develop an efficient and scalable solution.
- Applied principles of embedded systems, computer vision, and software-hardware integration.

Technologies: *Python, Embedded Systems, Facial Recognition, Computer Vision*

Smart Locker Using Facial Recognition · Final Year Project

2025 – Present

- Designed and developed a smart locker system using facial recognition technology for secure access control.
- Integrated camera-based authentication with embedded hardware components.
- Implemented user verification and access management functionality.
- Applied principles of embedded systems, computer vision, and software-hardware integration.

Technologies: *C/C++, ESP32/ESP8266, IoT, Embedded Systems, HTML, CSS, JavaScript, Sensors, MQTT/Web Technologie*

Systems & Integration Tracker · Firmlinx (Pty) Ltd

2026

- Developed and enhanced a web-based Systems & Integration Tracker platform.
- Implemented backend functionality using Flask and SQLAlchemy.
- Built integrations with external systems and third-party APIs.
- Developed dashboards, reporting tools, user assignment workflows, notifications, and file management features.

  Technologies: *Python, Flask, SQLAlchemy, PostgreSQL, JavaScript, REST APIs*

Invoice management system · Freelance Project

2026

- Designed and developed an invoice management system to streamline the creation, management, tracking, and processing of business invoices..
- Implemented functionality for creating, updating, viewing, and managing invoices and customer information.
- Developed a structured database system for storing and managing invoice records, customer details, transactions, and related business data
- Integrated backend services and application logic to support reliable data processing and efficient invoice management.
- Applied software engineering principles, database management, Restful APIs, and web application development practices to build a scalable and user-friendly solution.

  Technologies: *Python, Flask, SQL, SQLite/PostgreSQL, REST APIs, HTML, CSS, JavaScript*

EDUCATION

Diploma in Computer Systems Engineering (Extended) · Tshwane University of Technology

Final Year | Expected Graduation: 2026 | Pretoria, South Africa

- Specializations: Hardware & Software Systems
- Digital Electronics & Electronic Circuits
- PLC Programming (Ladder Logic, Instruction List)
- Robotics & Automation Systems (ROS)
- Networking (TCP/IP, Routing, Switching)
- Operating Systems (Linux-based)
- Programming: C++, C#, Assembly (PIC Assembler), SQL
- Tools: Arduino, Proteus, Processing

Cambridge International AS & A Levels · British International College

2019

- Subjects: Mathematics, Accounting, Business Studies, English

LEADERSHIP & VOLUNTEER EXPERIENCE

Class Representative – Mobile Robotics · Tshwane University of Technology

July 2024 – November 2024

- Bridged communication between faculty and students.
- Organized peer study groups and conveyed academic feedback.
- Assisted classmates with academic and project-related concerns.

INTERESTS

- Software Development and System Integration
- Cloud Computing and Cyber Security
- Robotics and Automation Technologies
- Artificial Intelligence and Machine Learning
- Electrical System Innovation and Troubleshooting
- Mentoring Peers and Community Development

REFERENCES

Name	Phone number	Email
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